Defence Run Sheet

Interactive Web GIS Portal for Flood Risk Analysis, Lagos State | Attah Gbubemi David

The ten numbers to know cold

4,810,069	**38.7%**	**0.0115**	**0.507**	**0.57**
people in High or Very High zones	of the Lagos population	AHP consistency ratio (limit 0.10)	SWAT+ discharge NSE, full period	validation NSE (calibration 0.41)
19,915	**23 / 243**	**0.62**	**1.61**	**3,490**
km^2 delineated Ogun catchment	subbasins / HRUs	map AUC-ROC (target 0.70)	flood enrichment, Very High zone	mm 100-year design rainfall

Plus: Eti-Osa 67.8% exposed (worst LGA), Ifako/Ijaye 7.2% (best). +10% rainfall gives +15.5% discharge.

Before you start

Run every launcher once the night before. Click every portal tab. Close Chrome, Brave, Edge, Zoom and Docker before opening QGIS — memory pressure is what crashed it during preparation.

Open this	From	Why
Portal	1_run_portal.bat → localhost:8501	the deliverable
Hazard map project	C:\SWAT_Ogun\OgunModel2\lagos_demo.qgz	proof you built the map
SWAT+ project	C:\SWAT_Ogun\OgunModel2\OgunModel2.qgs	the catchment model
AHP workbook	AHP_weights.xlsx	live weights, not a picture
Lab notebook	flood_model_bundle\swat\METHODOLOGY_LOG.md	the record of failures

Note: lagos_demo.qgz saved into the OgunModel2 folder, not the Desktop.

The sequence — about 20 minutes

#	What	Time	The line that matters
1	Portal, 8 tabs	6 min	the finished product, working
2	QGIS: lagos_demo.qgz	3 min	“these are my intermediates”
3	6_validate_my_rebuild.bat	90 s	AUC difference 0.0000
4	AHP_weights.xlsx	1 min	type 9 into B14, then Ctrl+Z
5	QGIS: OgunModel2.qgs	3 min	23 subbasins, outlet at Kara
6	3_run_swat_plus.bat	30 s	14 years of daily water balance
7	4_recheck_calibration.bat	40 s	NSE re-derived, not recalled
8	METHODOLOGY_LOG.md	2 min	the failures nobody could fake

1. The portal — eight tabs

Browser at **localhost:8501**. Spend your time on Point Query; move briskly through the rest.

Overview — 20 seconds

> *"Scope statement. This is a susceptibility and scenario tool, not a flood forecast. It shows where flooding is likely, not when one will happen."*

Hazard Map — 45 seconds

> *"Five classes from the AHP weighted overlay, with the twenty LGA boundaries over it. Red follows the lagoon, the Ogun outlet and the creek network. That is the model responding to elevation and wetness index, which together carry 0.38 of the weight."*

Point Query — 90 seconds • your best minute in the whole demo

Click two contrasting presets and read the contrast aloud.

Preset	Zone	Elevation	Slope	TWI	Runoff
Lagos Island	High	6.8 m	1.88°	7.57	1402 mm
Ikeja (Mainland)	Moderate	35.0 m	0.63°	13.93	1353 mm

> *"Same model, two points. Lagos Island sits at 6.8 metres and comes out High. Ikeja is at 35 metres and comes out Moderate, even though its wetness index is nearly double, because elevation carries more weight. That is the AHP weighting working, not a lookup table."*

Exposure — 45 seconds

> *"4.81 million people in High or Very High, 38.7% of the state. Eti-Osa is worst at 67.8%, Ifako/Ijaye lowest at 7.2%."*

Scenario — 60 seconds

Slider to **+30%**, then **Run Scenario**.

> *"Rainfall drives only the runoff and extreme-rainfall factors, 0.20 of the index, so the map shifts modestly. That is correct, not a bug: Lagos flood hazard is topography-structured. The process-based answer is in the Hydrology tab, where SWAT+ turns a 10% rainfall increase into 15.5% more discharge."*

Climate — 45 seconds

Type **100** into return period → 3,490 mm. Then type **3932** into annual rainfall → about 289 years.

> *"3,932 millimetres was actually observed in 2018. On this Gumbel fit that is a 289-year event, which tells you the distribution understates the upper tail at this station. I report that as a limitation."*

Hydrology (SWAT+) — 60 seconds

> *"23 subbasins, 19,915 square kilometres, 243 HRUs. Validation NSE 0.57 against calibration 0.41. Validation scoring higher than calibration means it is not overfitted. And plus 10% rainfall gives plus 15.5% flow. That non-linearity is the whole reason SWAT+ was worth building alongside the index."*

Validation — 45 seconds

> *“AUC 0.62 against a 0.70 target. The map ranks hazard correctly, with enrichment rising monotonically to 1.61 in the Very High zone, but it does not discriminate sharply. MODIS at 250 metres cannot see street-scale urban flooding.”*

2. The hazard map project — lagos_demo.qgz

Scroll the Layers panel **bottom to top**. That is build order.

> *"Bottom group: the eight conditioning factors, each rescaled to nought to one so they are comparable. Three are inverted — elevation, slope and distance to river — because low, flat and near water is the dangerous end. dist_n I had to rebuild; my first expression produced infinities. sum1 and sum2 are the weighted sum in two halves, four factors each, split because the Raster Calculator ran out of memory on twelve layers at once. mask is the validity rule: a cell only gets an index if all eight factors have data there. That is why my total is 4.2 million cells, not the full 7.4 million rectangle. FSI is the index itself. hazard_zones is the five classes. That is Figure 8."*

Then the point of showing it at all:

> *"These intermediates are the argument. A finished map can be copied. Half-finished weighted sums that exist because a piece of software crashed cannot."*

3. Prove the rebuild — 6_validate_my_rebuild.bat

Measure	Published map	Your rebuild
Valid cells	4,202,132	4,202,265
AUC-ROC	0.620	0.620
Sensitivity	0.685	0.684
High+V.High enrichment	1.37	1.38
Cell-for-cell agreement	—	95.92%
Within one class	—	100.00%

> *"I rebuilt the published map from the raw conditioning factors. 133 cells different out of 4.2 million, and not one cell in the state is more than a single class away. The AUC agrees to three decimal places, which means I reproduced the model, not just the picture."*

4. The weights are mine — AHP_weights.xlsx

Point at the blue block, then at CR = 0.0115. Then type **9** into **B14**: CR jumps to 0.3172 and the verdict flips to REJECTED. **Ctrl+Z**.

> *"28 pairwise judgements are the only input. The weights come from the geometric mean of each row. The consistency ratio asks whether my judgements contradict each other. At 0.0115 they are about one percent as inconsistent as filling the matrix in at random. And it calculates — it is not a picture of a spreadsheet."*

5. The catchment model — OgunModel2.qgs

> *“This is the watershed delineation, the spatial skeleton of the model. QSWAT+ ran TauDEM over the 90-metre DEM with the river network burned in and produced 23 subbasins totalling 19,915 square kilometres, which matches the published extent of the Ogun. The headwaters are at about 8.9 degrees north in Oyo State; everything drains south. Subbasin 23 is the outlet at Kara/Akute, just above where the Ogun enters the Lagos Lagoon. That single point is what the whole model is calibrated against. Inside these 23 subbasins there are 71 channels and 243 hydrologic response units, each a unique combination of land use, soil and slope.”*

Optional: click the **Identify** tool, then any subbasin, to show its attributes.

6. Run the engine — 3_run_swat_plus.bat

Each scrolling line is one simulated day, marching 2010 to 2023.

> *"This is the SWAT+ engine, revision 62, the actual executable. For each of 243 hydrologic response units it solves a daily water balance — interception, infiltration, runoff, percolation, evapotranspiration, baseflow — then routes it through 71 channels to the outlet. Fourteen years in thirty seconds. And this timestamp is the file it just wrote: ten megabytes of monthly discharge. That is the file my calibration statistics are computed from."*

7. Re-score it — 4_recheck_calibration.bat

Variable	NSE	R^2	PBIAS	Threshold met
Discharge vs GEOGLOWS	0.507	0.606	+1.6%	all three — PASS
Evapotranspiration vs satellite	0.180	0.394	+12.5%	volume only

> *"This writes my calibrated parameter set, runs the engine again, and scores it against two independent variables at once. Discharge meets all three Moriasi thresholds, and the script checks that itself rather than me claiming it. The second line is the part I would point to: evapotranspiration was never a calibration target, so scoring it against satellite data is an independent test of whether the water balance is right internally, not just at the outlet."*

8. The lab notebook — METHODOLOGY_LOG.md

Scroll to the troubleshooting entries and name three:

What went wrong	How it was diagnosed and fixed
TauDEM D8FlowDir hung 3.4 hours	Watched the output folder and CPU time; killed it; moved from 30 m to 90 m DEM
Flow accumulation maxed at 462 km^2 on a 23,000 km^2 basin	DEM verified clean, so a routing problem: burned in HydroRIVERS, moved the outlet 2.3 km west; result 20,592 km^2
HRU build failed, "Id 490 not in gis_lsus"	Degenerate micro-channel from burn-in; raised channel threshold 40 to 150 km^2, channels 277 to 71

> *"This is the record of it going wrong."*

Success can be copied. A three-and-a-half-hour hang with the diagnosis attached cannot.

How each stage was made — one sentence each

Stage	Say this
Grids	Everything reprojected to UTM 31N and resampled to one 6053 x 1225 grid at 30 m — bilinear for continuous surfaces, majority for land cover, nearest for the curve-number lookup.
Runoff	SCS curve number. $S = 25400/CN - 254$, then $Q = (P - 0.2S)^2 / (P + 0.8S)$.
Flow accumulation	Log-transformed before normalising — raw values span seven orders of magnitude, so without it every cell but the channels sits at zero.
Land cover	Not measured but scored, built-up 0.90 down to water 0.10, from the literature. The most subjective input, which is why it carries the smallest weight at 0.05.
Normalisation	Min–max on the 2nd to 98th percentile, with elevation, slope and distance to river inverted.
Weights	28 pairwise comparisons, geometric mean, consistency ratio 0.0115 against a 0.10 threshold.

Stage	Say this
Overlay	One weighted sum of eight normalised factors, then five equal-interval classes at 0.20.
Validation	MODIS 250 m observed flood extent for Lagos, July to November 2012, from NASA/DFO via HDX.

Questions to expect

Are the two models connected? Do they combine?

No, and that was deliberate. The AHP map is statewide and static — it says where in Lagos is susceptible. SWAT+ is a single catchment and dynamic — it says how much water the Ogun delivers and how that responds to rainfall. They share source datasets but no outputs. Substituting a single-catchment discharge into a statewide index would be unsound, so the runoff factor is an SCS Curve Number estimate computed across the whole state, and SWAT+ sits in its own panel. That is stated in Section 3.6.

Then why build SWAT+ at all?

Non-linearity. Plus 10% rainfall gives plus 15.5% discharge, because once the soil saturates the extra rain has nowhere to go. A weighted linear index can only ever respond linearly. And they are two independent validations — one against observed flood extent, one against discharge — rather than one coupled chain where an error at either end propagates through.

Why not couple them properly?

Coupling means converting simulated discharge into inundation extent, which requires a hydraulic model routing a water surface across a floodplain. SWAT+ does not do that. Detailed hydraulic modelling is excluded in Section 1.7 and named as further work.

Why the Ogun basin and not Lagos?

SWAT models a drainage basin, not an administrative boundary. The water that floods Lagos through the lagoon falls as rain upstream and arrives down the Ogun. Modelling the Ogun basin is modelling Lagos's fluvial flood source. Oyan Dam sits in that basin and regulates flow, which I record as a limitation — it is why the 2022 flood does not appear in a rainfall-only simulation.

Who made the AHP judgements? Just you?

Yes, informed by the flood-susceptibility literature for Lagos and comparable tropical coastal cities. A group exercise with LASEMA and state planners would be stronger and I identify it as further work. The consistency ratio establishes internal coherence, not external authority.

A low consistency ratio just means you were careful, not right.

Correct, and the distinction matters. CR tests consistency, not correctness — a perfectly consistent set of wrong judgements would score zero. That is why the weighting is not the only check: the resulting map is validated independently against the MODIS 2012 flood extent. The spreadsheet tests the reasoning; the flood extent tests the result.

NSE 0.507 against a threshold of 0.50 is very close.

It is, and I flag it in my methodology log. The margins are thin on both NSE and R-squared, which is why I re-run the check after any change rather than trusting a recorded number. The split-sample result matters more: validation on 2018 to 2023 scores 0.57, higher than calibration at 0.41, indicating the fit is not overtuned.

Why not improve the evapotranspiration?

I tried. Raising petco improves ET monotonically but degrades discharge — at 1.35 the ET NSE reaches 0.30 while R-squared drops below 0.60 and fails. The objectives genuinely conflict, and petco is already at its permitted maximum of 1.20. Two unrelated satellite products both say the model wants more ET than the parameter range allows.

How did you get 4.81 million exposed?

Zonal statistics — population summed per hazard zone, using native-resolution WorldPop in the original pipeline. The copy in the deployment bundle is a 30-metre resample kept only for scenario ratios; it inflates counts about eightfold, which is why the portal reads the validated figures from file rather than recomputing them.

Did you do every step by hand?

I built the model and the layers. The final combination was run as a script using identical expressions, because the Raster Calculator exhausted memory on a twelve-layer expression on this machine. The script and the QGIS route produce the same file, and the validation batch file verifies it against the published map.

Four things to say before anyone asks

The weakness	Own it like this
AUC 0.62 misses the 0.70 target	The map ranks hazard correctly but does not discriminate sharply. MODIS at 250 m cannot see street-scale urban flooding, and one 2012 event is being tested against a static surface. It is a valid screening tool for prioritising LGAs, not a substitute for hydraulic modelling.
ET seasonality is about 3x too flat	SWAT+ swings 26 mm across the year against MODIS's 86 mm. It is the single most defensible criticism of the model, and petco is already pinned at its maximum.
The NEMA 2022 check is inconclusive	Report counts track raw population at rho +0.57 far better than they track hazard at +0.17. The dataset measures reporting effort, not susceptibility. I report it as a null result, neither support nor refutation.
Chapter 1 says “SWAT-derived runoff indicators”	That is the proposal's wording. Section 3.6 states explicitly that the runoff factor is SCS-CN, not SWAT+ output, and explains why. Get there before anyone else does.